La Scuola d'Italia / Huson / Physics Nickname:
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2.6 Warm-up: Mental Math (no calculator)

Work each question in your head — no calculator, and no need to show work. Write just the answer, with its units.

Reference. kilo (k) = 1000 centi (c) = $\frac{1}{100}$ milli (m) = $\frac{1}{1000}$ 1 m/s = 3.6 km/h

1. Fill in the blanks.

   (a) 3.2 km = _____ m

   (b) 60 cm = _____ m

   (c) 1.5 m = _____ cm

   (d) 9 mm = _____ cm

   (e) 0.4 m = _____ mm

   (f) 750 m = _____ km

2. Squares and square roots.

   (a) 4^2 = _____ (c) 11^2 = _____ (e) 0.2^2 = _____ (g) $\sqrt{49}$ = _____

   (b) 9^2 = _____ (d) 30^2 = _____ (f) 1.5^2 = _____ (h) $\sqrt{400}$ = _____

3. Use 1 m/s = 3.6 km/h.

   (a) 5 m/s = _____ km/h

   (b) 15 m/s = _____ km/h

   (c) 72 km/h = _____ m/s

   (d) 90 km/h = _____ m/s

4. Distance = speed × time.

   (a) A student walks at 1.5 m/s for 20 s. How far? _____

   (b) A cyclist rides at 6 m/s for one minute. How far? _____

   (c) A car drives at 90 km/h for 20 minutes. How far? _____

   (d) How long does it take to run 100 m at 5 m/s? _____

5. A runner jogs one lap of a 400 m track in 80 s, finishing where the lap began.

   Average speed = _____ Average velocity = _____

6. Write each fraction as a decimal.

   (a) $\frac{3}{5} =$ _____ (c) $\frac{3}{20} =$ _____ (e) $\frac{3}{25} =$ _____ (g) $\frac{1}{9} =$ _____

   (b) $\frac{9}{10} =$ _____ (d) $\frac{1}{50} =$ _____ (f) $\frac{5}{4} =$ _____ (h) $\frac{4}{3} =$ _____

7. Fill in the blanks.

   (a) 0.5 hour = _____ min

   (b) 0.4 hour = _____ min

   (c) 2.5 min = _____ s

   (d) 90 s = _____ min

   (e) 6 min = _____ h (decimal)

   (f) 40 min = _____ h (fraction)

   (g) 1 week = _____ hours

   (h) 2.25 hours = _____ h _____ min

8. Acceleration = change in velocity ÷ time.

   (a) A car goes from 0 to 20 m/s in 5 s. Find its acceleration. _____

   (b) A bike speeds up from 2 m/s to 8 m/s in 3 s. Find its acceleration. _____

   (c) A car at 20 m/s brakes, losing 5 m/s every second. How many seconds does it take to stop? _____

   (d) For the car in (a): average velocity = _____ distance in the 5 s = _____

9. Use $g \approx 10$ m/s^2: a falling object gains 10 m/s of speed every second.

   (a) A stone is dropped. How fast is it going after 3 s? _____

   (b) How far has it fallen after 2 s? $(\frac{1}{2} \times 10 \times t^2)$ _____

   (c) A ball is thrown straight up at 20 m/s. How many seconds does it take to reach the top? _____

   (d) At the top: velocity = _____ acceleration = _____

10. A marathon is about 42 km, and 1 km $\approx \frac{5}{8}$ mile. About how many miles? _____